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Code No: 117FZ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, February/March - 2022
OPERATIONS RESEARCH
(Common to ME, CSE, MIE, AGE)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. Solve the following L.P.P. by using simplex method:
   Maximize $Z = 4x_1 + 10x_2$
   Subject to $2x_1 + x_2 \leq 50$
   $2x_1 + 5x_2 \leq 100$
   $2x_1 + 3x_2 \leq 90$
   and $x_1, x_2 \geq 0$ [15]

2. A Person requires at least 4 and 7 units of chemicals A and B respectively for his garden. A typical liquid product contains 2 and 1 unit of A and B, respectively per jar. On the other hand, a typical dry product contains 1 and 7 units of A and B per unit. If the liquid product sells for Rs. 1 per jar and the dry product for Rs. 1 per unit, how many of each should be purchased in order to minimize the cost and meet the requirement. Solve the L.P.P by using Big– M method. [15]

3. A product is manufactured at four factories A, B, C and D. Their production capacities are 50, 70, 30 and 50 units, respectively. The factories supply the product to four stores (I, II, III and IV) demand of which are 25, 35, 105 and 20 units respectively. Unit transportation cost in rupees from each factory to each store is given in the table below:

	I	II	III	IV
A	4	6	8	13
B	13	11	10	8
C	14	4	10	13
D	9	11	13	8

   Determine the extent of deliveries from each of the factories to each of the stores, so that total transportation cost is the minimum. [15]

4. A firm produces four products. There are four operators who are capable of producing each of them. The processing time varies from operator to operator. The processing times in minutes of each of the product are given below. Find the optimal assignment of products to operators. [15]

	Products			
Operators	A	B	C	D
1	10	14	22	12
2	16	10	18	12
3	8	14	20	14
4	20	8	16	6

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5. There are 5 projects to be made, each of which must undergo three stages viz., information feeding, analyzing and printing. A programmer has decided to operate three computers namely zebronics, Wipro and Acer for these operations. Processing times are given below:

	Projects	P	Q	R	S	T
Computers	Acer	8	5	4	6	5
	Zebronics	10	13	11	8	12
	Wipro	6	2	9	7	4

Determine the total elapsed time and idle time of each computer. [15]

6. A machine A costs Rs.9000/-. Annual operating costs are Rs. 200/- for the first year and then increases by Rs.2000/- every year. Determine the best age at which the machine A is to be replaced? If the optimum replacement policy is followed, what will be the average yearly cost of owning and operating the machine? Machine B costs Rs. 10000/-. Annual operating costs are Rs. 400/- for the first year and then increases by Rs. 800/- every year. You have now a machine of type A, which is of one year old. Should you replace it with B, and if so, when? Assume both machines have no resale value when replaced and that future costs are not discounted. [15]

7. A shopkeeper has a uniform demand of an item at the rate of 50 items per month. He buys from a supplier at a cost of Rs.6/- per item and the cost of ordering is Rs. 10/- per order. If the stock holding costs are 20% of stock value, how frequently should he replenish his stock? Suppose the supplier offers 5% discount on orders between 200 and 999 items and a 10% discount on orders exceeding or equal to 1000 units. Can the shopkeeper reduce his costs by taking advantage of either of these discounts? [15]

8. A tax consulting firm has 3 counters in its office to receive people who have problems concerning their income, wealth and sales taxes. On the average 48 persons arrive in an 8-hour day. Each tax adviser spends 15 minutes on an average on an arrival. If the arrivals are poissionly distributed and service times are exponential distributed, find (a) The average number of customers in the system (b) Average waiting time for a customer (c) Average number of customers waiting to be serviced (d) Average time a customer spends in the system. [15]

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